3-3

(1) 气体可以当作是理想气体，理想气体的内能是温度的单值函数。

选取绝热气缸内的两部分气体共同作为热力学系统。

在过程中，气体与外界没有热量交换（绝热气缸），对外也不作功（刚性气缸）

故内能变化 $\Delta U=0 \Rightarrow$ 系统终温 = 初温 = 30°C

(2) $p_1V_1 = m_1R_gT_1$

$$V_1 = \frac{m_1R_gT_1}{p_1} = \frac{0.5\times287\times(273+30)}{0.4\times10^6} = 0.1087\,m^3$$

$$V_2 = \frac{m_2R_gT_2}{p_2} = \frac{0.5\times287\times303}{0.12\times10^6} = 0.3623\,m^3$$

$$V_{Total} = V_1+V_2 = V_1'+V_2' = 0.471\,m^3$$

当系统达到温态后，$p_1'=p_2'=p$

故 $\frac{p_1V_1}{T_1} = \frac{p_1'V_1'}{T_1'} = \frac{pV_1'}{T_1}$ $\qquad \frac{p_2V_2}{T_2} = \frac{p_2'V_2'}{T_2'} = \frac{p(V_{Total}-V')}{T_2}$

$43480 = pV_1'$

$$p = \frac{43480}{V_1'}$$

$$p = \frac{43476}{V_{Total}-V_1'} = \frac{43480}{V_1'}$$

$(0.471-V_1')(43480) = 43476V_1'$

$20479.08 - 43480V_1' = 43476V_1'$

$V_1' = 0.2355$

$p = 1.8463\times10^5\,Pa$

3-5

活塞和氢气所处气缸是绝热的，所以氢气在过程中没有从外界吸入热量

∴ 可看作可逆绝热过程

$$\frac{T_{H_2'}}{T_{H_2}} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} \qquad T_{H_2'} = T_{H_2}\left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} = (15+273)\left(\frac{1.9614}{0.9807}\right)^{\frac{1.41-1}{1.41}}$$

$= 352.3109\,K$

$$V_{H_2}' = \frac{P_{H_2} V_{H_2} T_{H_2}'}{P_{H_2}' T_{H_2}} = \frac{0.9807\times10^5 \times 0.1 \times 352.3109}{1.9614\times10^5 \times 288} = 0.0612\,m^3$$

$$V_{O_2}' = 0.2 - 0.0612 = 0.1388\,m^3$$

$$T_{O_2}' = \frac{P_{O_2}' V_{O_2}' T_{O_2}}{P_{O_2} V_{O_2}} = \frac{1.9614\times10^5 \times 0.1388 \times 288}{0.9807\times10^5 \times 0.1} = \underline{799.488\,K}$$

$$Q = \Delta U = \Delta U_{O_2} + \Delta U_{H_2} = m_{O_2} c_{V_{O_2}} (T_{O_2}' - T_{O_2}) + m_{H_2} \frac{1}{k-1} R_g (T_{H_2}' - T_{H_2})$$

$$R_{g(O_2)} = \frac{8.3145}{0.029} = 286.7069 \approx 287$$

$$= \frac{P_{O_2} V_{O_2}}{R_{g(O_2)} T_{O_2}} c_{V_{O_2}} (T_{O_2}' - T_{O_2}) + \frac{P_{H_2} V_{H_2}}{R_{g(H_2)} T_{H_2}} R_g (T_{H_2}' - T_{H_2})$$

$$R_{g(H_2)} = \frac{8.3145}{0.002} = 4157.25 \approx 4157$$

$$= \frac{0.9807\times10^5 \times 0.1}{287\times288} (0.71594)(799.488 - 288)$$

$$+ \frac{0.9807\times10^5 \times 0.1}{4157\times288} \left(\frac{1}{1.41-1}\right)(352.3109 - 288)$$

$$= \underline{44.7333\,J}$$

3-7

(1) 定温 $T_1 = T_2 = \underline{303K}$

$$V_1 = \frac{mR_gT_1}{P_1} = \frac{6\times287\times303}{0.3\times10^6} = 1.73922\,m^3$$

$$V_2 = \frac{mR_gT_2}{P_2} = \frac{6\times287\times303}{0.1\times10^6} = 5.21766\,m^3$$

$$W = \int_{V_1}^{V_2} p\,dV = mR_gT_1 \ln\frac{V_2}{V_1} = 6\times287\times303\times\ln\frac{5.21766}{1.73922} = \underline{573.22\,kJ}$$

$$Q = -W = \underline{-573.226\,J}$$

(2) 定熵：可逆绝热

$$W=\int_{V_1}^{V_2}p\,dV=\frac{k}{k-1}p_1V_1\left[1-\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}\right]$$

$$=\frac{1.4}{1.41-1}\times 6\times 287\times 303\left[1-\left(\frac{1}{3}\right)^{\frac{1.41-1}{1.41}}\right]=\underline{490.678kJ}$$

$$\frac{T_2}{T_1}=\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}\qquad T_2=T_1\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}=303\left(\frac{1}{3}\right)^{\frac{1.41-1}{1.41}}=\underline{220.1432K}$$

$$Q=\underline{0kJ}$$

(3) $n=1.2$ 多方过程

$$T_2=T_1\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}=303\left(\frac{1}{3}\right)^{\frac{1.2-1}{1.2}}=252.303K$$

$$W=\frac{mR_gT_1}{n-1}\left[1-\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}\right]=\frac{6\times 287\times 303}{1.2-1}\left[1-\left(\frac{1}{3}\right)^{\frac{1.2-1}{1.2}}\right]$$

$$=\underline{436.5011kJ}$$

$$Q=mc_v(T_2-T_1)\frac{n-k}{n-1}=6\times 0.717\times(252.303-303)\left(\frac{1.2-1.41}{1.2-1}\right)$$

$$=\underline{229.0034kJ}$$

3-15

(1) 空气是双原子分子

$$c_v=\frac{5}{2}\cdot\frac{R}{M}=\frac{5}{2}(8.3145)\left(\frac{1}{28.97}\right)=0.7175\ kJ/kg\cdot K$$

$$c_p=\frac{7}{2}\cdot\frac{R}{M}=\frac{7}{2}(8.3145)\left(\frac{1}{28.97}\right)=1.0045\ kJ/kg\cdot K$$

可逆绝热过程方程：

$$p_2 = p_1\left(\frac{T_2}{T_1}\right)^{\frac{k}{k-1}} = 0.1\times10^6\left(\frac{480}{300}\right)^{\frac{1.41}{0.41}} = 0.5035\,\text{MPa}$$

$$\Delta u = c_V\Delta T = 0.7175(180) = \underline{129.15\ \text{kJ/kg}}$$

$$w = -\Delta u = \underline{-129.15\ \text{kJ/kg}}$$

(2) 查表可得 $t_1 = 27°C = 300K$, $u_1 = 214.32$ kJ/kg

$t_2 = 207°C = 480K$, $u_2 = 345.04$ kJ/kg

$$\Delta u = u_2 - u_1 = \underline{130.72\,\text{kJ/kg}}$$

$$w = -\Delta u = \underline{-130.72\,\text{kJ/kg}}$$

3-20

(1) $$R_g = \frac{pV}{mT} = \frac{0.2\times10^6\times2}{5\times283.69} = \underline{282.0\ \text{J/kg}\cdot\text{K}}$$

$$M = \frac{R}{R_g} = \frac{8.3145}{0.282} = 29.484$$

ratio of $O_2 = \frac{2}{5}$ ratio of unknown gas $= \frac{3}{5}$

(2) $$n = \frac{pV}{RT} = 169.582\ \text{mol}$$

$$n_{O_2} = \frac{m}{M_{O_2}} = \frac{2000}{32} = 62.5\,\text{mol}$$

未知气体分压力 $= \frac{n - n_{O_2}}{n} = \frac{169.582 - 62.5}{169.582}(0.2\times10^6) = \underline{0.1263\ \text{MPa}}$

3-21

$$s_2 - s_1 = \int_1^2 \frac{c_V}{T}dT + R_g\ln\frac{V_2}{V_1}$$

若理想气体比热容为常数

$$s_2-s_1=c_V\ln\frac{T_2}{T_1}+R_g\ln\frac{V_2}{V_1}$$

多方过程：

$$c_V=\frac{R_g}{k-1}\qquad \frac{V_2}{V_1}=\left(\frac{p_1}{p_2}\right)^{\frac{1}{n}};\quad \frac{T_2}{T_1}=\left(\frac{p_1}{p_2}\right)^{\frac{1-n}{n}}$$

$$s_2-s_1=\frac{R_g}{k-1}\ln\left(\frac{p_1}{p_2}\right)^{\frac{1-n}{n}}+R_g\ln\left(\frac{p_1}{p_2}\right)^{\frac{1}{n}}$$

$$=\frac{R_g(k-n)}{n(k-1)}\ln\frac{p_1}{p_2}$$

$$=\frac{(n-k)R_g}{n(k-1)}\ln\frac{p_2}{p_1}$$

又 $\frac{V_2}{V_1}=\left(\frac{T_1}{T_2}\right)^{\frac{1}{n-1}}$

$$s_2-s_1=\frac{R_g}{k-1}\ln\frac{T_2}{T_1}+R_g\ln\left(\frac{T_1}{T_2}\right)^{\frac{1}{n-1}}$$

$$=\frac{n-k}{(n-1)(k-1)}R_g\ln\frac{T_2}{T_1},\quad n\neq 1$$

3-25

$$p_1V_1=nRT_1$$

$$nT_1=\frac{p_1V_1}{R}=\frac{0.517\times10^6\times0.142}{8.3145}=8829.635\ \mathrm{K\cdot mol}$$

$$nT_2=\frac{p_2V_2}{R}=\frac{0.172\times10^6\times0.274}{8.3145}=5668.17\ \mathrm{K\cdot mol}$$

$$C_{p,m}=\frac{\Delta H}{n\Delta T}=\frac{-65.4}{(5668.17-8829.635)\times10^{-3}}=20.6866\ \mathrm{J/K\cdot mol}$$

$$C_{V,m}=C_{p,m}-R=20.6866-8.3145=12.3721\ \mathrm{J/K\cdot mol}$$

$$M = \frac{C_{v,m}}{C_v} = \frac{12.3721}{1.41} = 8.7745 \text{ g/mol}$$

$$\Delta U = nC_{v,m}\Delta T = 12.3721 \times (5668.17 - 8829.635) = \underline{-39.114 \text{ kJ}}$$

$$C_p = \frac{C_{p,m}}{M} = \underline{2.3576 \text{ kJ/kg·K}}$$